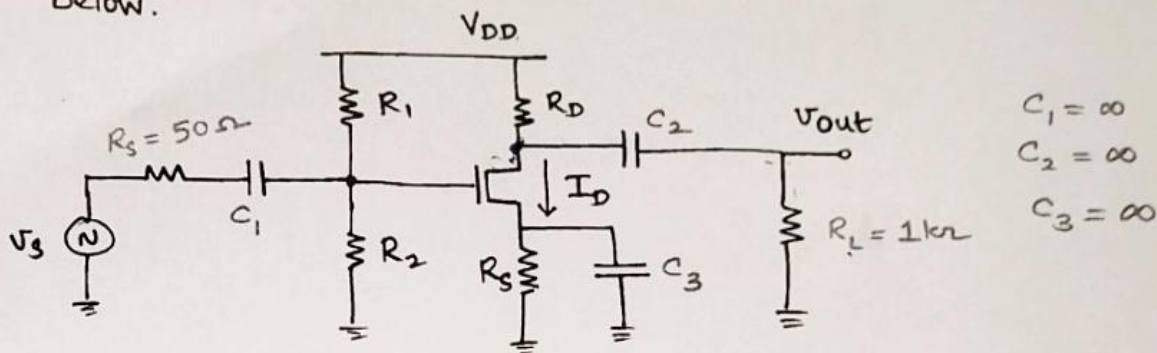


- ① Calculate the voltage gain of the common-source amplifier below.



$$V_{DD} = 3V, R_2 = 10k\Omega, R_1 = 20k\Omega, R_D = 1k\Omega, R_S = 0.2k\Omega$$

Assume the MOSFET parameters as $\mu_n C_{ox} = 400 \mu A/V^2$, $(\frac{W}{L}) = 500$, $V_{Th} = 0.7V$, $\lambda = 0$ and the transistor is in saturation.

- ② If $v_s = 0.05 \cos[2\pi(10000)t]$. Calculate the power efficiency of the common source amplifier.

- ③ In problem-2, due to non-availability of a MOSFET, an NPN BJT needs to be used. The BJT parameters are given below.

$$I_S = 10^{-16} A, V_T = 25mV, \eta = 1, \beta = 200, |V_A| = \infty$$

- (a) The quiescent current through the BJT needs to be the same as in problem-2. What should the new value of R_2 in this case. Ignore base current (I_{BQ}) for quiescent calculations.
- (b) Calculate the small signal voltage gain of the common emitter amplifier. [You need the solution of problem-1 for this]

④ If $v_s = 0.05 \cos [2\pi(10000)t]$. Calculate the power efficiency of the common emitter amplifier of Problem-4. Compare this with that of the common source amplifier, calculated in problem-3.

⑤ Consider the circuit of problem-2. A student forgot to solder the capacitor C_3 . What is the new voltage gain because of this?